DARK

ANALYST REQUEST

which device is bound to ACCT-4cd0?

LOG STORE

19,838 tokens
6 entities on every line

THE CONFLICT

redact → no handle
compress → 88.8–100% leak

CLOUD READER

receives (q', P)
one short answer

I. DATA EXAMPLE

··· other records ···

```
media_player
ACCT-4cd0
temp 21.4
```

··· other records ···

same block
in every case
multi / needle

II. NO SAFE CHOICE

two failures, one cause

0.1712	0.4831
mask	Quito-X
0%	98.1%

leak rate

III. THE RULE

Delete — **A**lias
Refill — **K**eep

two labels per unit,
four operations, no training

IV. DARK

DARK	**0.4937**
Quito-X	0.4831
CPC	0.4740
Provence	0.4080
LLMLingua-2	0.2651

vs Quito-X: tie (n.s.)
leak **0%** vs 98.1%
others: 88.8-100%

V. WHERE IT STAYS HARD

0.3273	0.6970
multi	needle

toolmem_raw 0.263
every arm < 0.45

DARK matches the strongest compressor on F1 at zero measured leakage, on half the tokens.

6 corpora · 2,728 evaluation cases · 112 main · 3 × 300 public variants